

# Pipelined Processor Design

## 1.2.1 Datapath Design

### 1.2.1.1

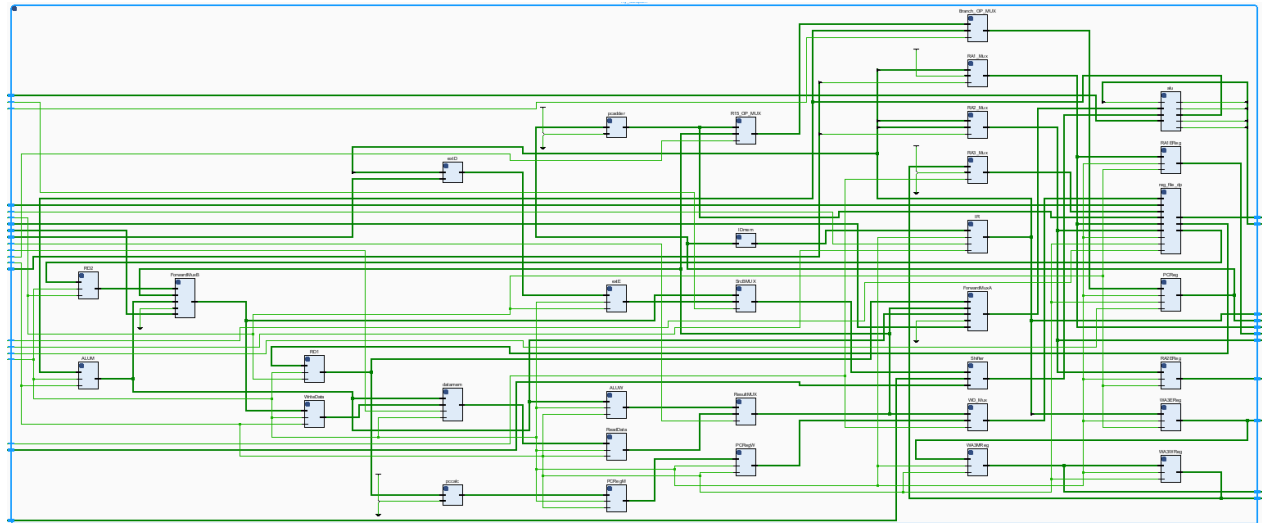


Figure 1 Synthesized Datapath's RTL view

### 1.2.1.2

As illustrated in Figure 1, a shifter has been inserted between the ALUSrcB multiplexer and the ALU's B input to support register-shifted immediates, rotated immediates, and MOV instructions. This shifter accommodates both shift types used in data-processing operations. It operates based on two control signals—shamt and shcontrol—which are produced by the controller and will be detailed in the controller section. The BX instruction does not require any extra hardware in the datapath; its execution is solely managed through control logic within the controller, which will also be covered later.

To support the BL (Branch with Link) instruction in the datapath, the signal linkW was introduced as a control signal that enables writing the return address into register R14 (link register). Specifically, when linkW is asserted, the destination register for the write-back stage is forced to R14 using a multiplexer (RA3\_Mux), and the data to be written is selected via another multiplexer (WD\_Mux) as PCW, which holds the return address (PC + 8). PC+8 value is obtained by RD1E – 4 in the execute stage and pushed through the write back stage. This allows the program to branch to the target while saving the return address for later use. The control and generation of the linkW signal itself is handled in the controller module.

## 1.2.2 Controller Design

### 1.2.2.1

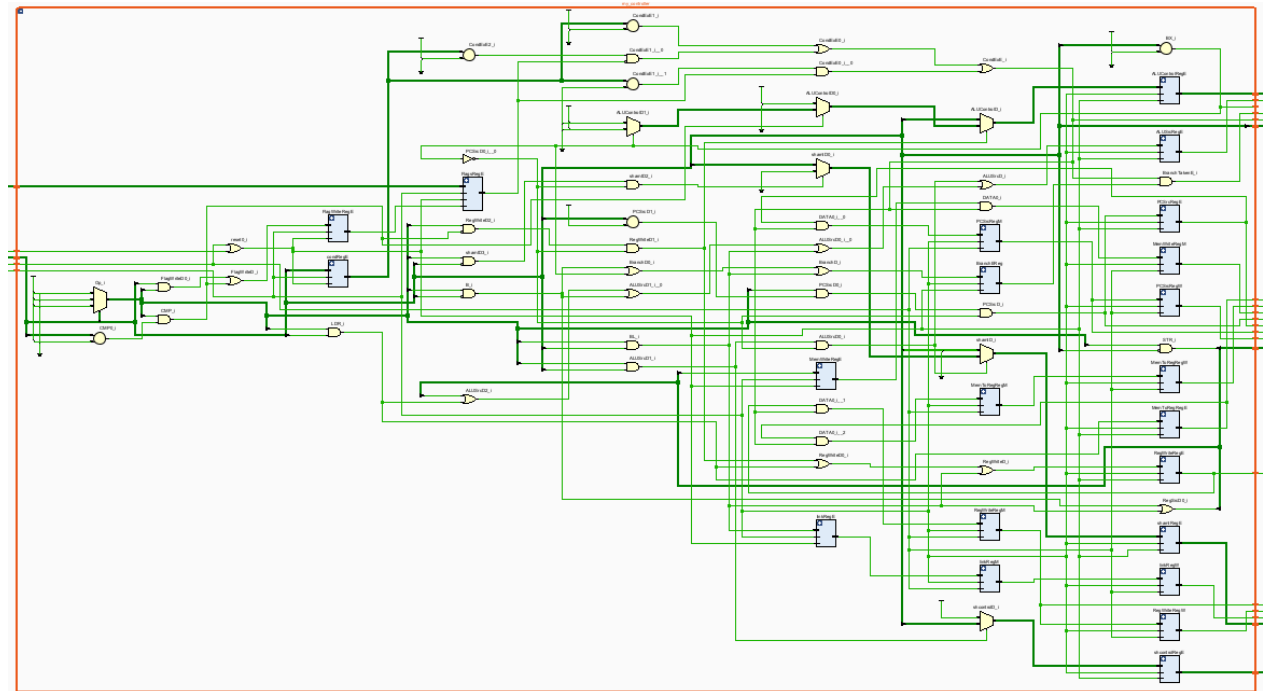


Figure 2 Synthesized Controller's RTL view

### 1.2.2.2

The shcontrol signal is derived based on the instruction type. For register-based data-processing instructions, it is taken directly from bits [6:5] of the instruction. In contrast, for immediate-based instructions, it is set to 2'b11 to indicate a rotate right operation. The shamt signal determines the shift amount and is also dependent on the instruction type. For register-based formats, it comes from bits [11:7], while for immediate formats, it is calculated as twice the value of bits [11:8], conforming to ARM encoding. In other instructions, shifting is disabled by setting shamt to zero. The controller classifies the instruction type each cycle to generate correct control signals. For the BX instruction, the ALU executes a MOV<sub>B</sub> to update the PC. Other branch instructions like B and BL perform an ADD to compute the new PC using  $PC + 8 + \text{immediate}$ .

The BL (Branch with Link) instruction is detected in the decode stage when  $Op == 2'b10$  and  $IROut[24] == 1$ . The controller asserts the link signal, which is passed through the pipeline stages (linkE, linkM, linkW). In the writeback stage, linkW indicates that the return address should be written to R14. This enables subroutine calls while maintaining pipeline compatibility and conditional execution.

### Figure 4 Test Results